Session 2 (Live demos): PubMed search, Zotero, and Google Scholar (optional)

Foundations of Clinical Research · Level 1 · Neudata · *Instructor script*

When	Online, Saturday 19:30–22:30. **Part A, PubMed:** segment 2.1 (0:10–0:50), slide "Live demo: searching PubMed for the running example" (about 0:30, 20:00). **Part B, Zotero:** segment 2.2 (0:50–1:10), slide "Live demo: from PubMed to Zotero to Word" (about 0:55, 20:25, just after the "Citation styles" slide). **Part C, Google Scholar:** optional extra slide
Length	Part A 10 minutes (Steps 1–5 plus Send to citation manager), Part B 5 minutes (Step 6), Part C 3 minutes only if time allows
Question	*In adults with septic shock in the ICU, does a restrictive fluid strategy, compared with usual care, reduce 28-day mortality?*
Goal	Participants see a PICO question become a documented, repeatable search, and watch the hits fall step by step
Participants	Online: watch the shared screen, note hit counts on their exercise sheet and suggest synonyms in the chat. They do **not** search during the demo

> **Golden rule on numbers.** Never promise a hit count. Say: *"Expect tens of thousands for one keyword and a few thousand for the full string. The numbers change every day as new papers are added."* Ask participants to write down **today's** numbers and the date. That is the lesson.

Before the session (the day before and during the break)

1. **Rehearse once, end to end**, on your own computer and network, sharing your screen in the video platform.
2. Open two browser tabs:

- `https://pubmed.ncbi.nlm.nih.gov/`
- `https://scholar.google.com/`

3. Set browser zoom to 125–150% so participants can read the shared screen on a laptop.
4. Paste the three blocks below into a text file you can copy from (typing long strings live is slow and error-prone).
5. **Offline fallback:** the day before, capture screenshots of each step below (search box, results count, Search details, filters, an abstract page, Send to → Citation manager, the Zotero import, the Word citation in both styles, Google Scholar results and "Cited by"). Put them, in order, in a short slide set or a folder, and send a copy to the second trainer so they can share it if your connection fails. Write the date and hit counts on each.
6. Do **not** show a personal My NCBI or Google account while sharing your screen.
7. **Reference manager:** install Zotero desktop (free, zotero.org) and its Word plug-in on the presenter laptop. Start with an empty test library called "Session 2 demo" and open a blank Word

document. Check once that a `.nbib` file imports (File → Import…). If Zotero cannot be installed, use the screenshots instead.

Strings to copy

Step 1: one keyword

```
septic shock
```

Step 2: the three PICO blocks

```
("Shock, Septic"[Mesh] OR "septic shock"[tiab] OR sepsis[tiab])
```

```
("Fluid Therapy"[Mesh] OR "fluid restrict*"[tiab] OR "restrictive fluid*"[tiab] OR "conservative fluid
```

```
("Mortality"[Mesh] OR mortality[tiab] OR death[tiab])
```

Step 3: the full string (all on one line in the search box)

```
("Shock, Septic"[Mesh] OR "septic shock"[tiab] OR sepsis[tiab]) AND ("Fluid Therapy"[Mesh] OR "fluid
```

Part A: PubMed (10 min), step by step

Go slowly: this audience has never seen PubMed's advanced features. After every step, pause 20–30 seconds so participants can write the number of hits on their sheet (Exercise 1).

Step 1: One keyword (1.5 min)

- **Do:** type `septic shock` in the PubMed box, then click **Search**.
- **Audience sees:** a results page, with the number of results above the list (tens of thousands).
- **Say:** *"This is everything that mentions septic shock. That is far too many to read. Write down the number and today's date."*
- **Point out:** the **sort** menu (Best match vs Most recent) and the **filters** column on the left (on a narrow screen it may be behind a "Filters" button).

Step 2: Build the blocks with OR, join them with AND (2.5 min)

- **Do (fast route, default):** paste the full string from Step 3 into the main box, then click **Search**.
- **Do (if ahead of time):** go to **Advanced**, run the **P**, **I** and **O** blocks one by one (they appear as #1, #2, #3 in **History and Search Details**), then combine them with `#1 AND #2 AND #3`.
- **Audience sees:** each block has many hits, but the combination has far fewer (expect a few thousand).
- **Say:** *"OR inside a block makes it wider, because every synonym counts. AND between blocks keeps only papers that talk about all three ideas."* Point back to the Venn slide if needed.
- **Ask the class (chat):** *"Which other words would authors use for restrictive?"* Add one or two good suggestions (e.g. `de-resuscitation[tiab]`) live and show the change in hits.

Step 3: Look at "Search details" (1.5 min)

- **Do:** on the Advanced page, in **History and Search Details**, click the arrow (›) next to the latest line to expand the details.
- **Audience sees:** how PubMed translated the search: the MeSH term with its narrower terms, the truncated words and the field tags.
- **Say:** *"This is the single best way to check your search. Copy this line into your proposal with the date."*
- **Teaching point:** the asterisk finds all endings, `[tiab]` means title/abstract, and `[Mesh]` also searches the narrower headings below it.

Step 4: Add filters (2 min)

- **Do:** on the results page, left-hand filters:
- **Article type:** tick *Randomized Controlled Trial*, *Systematic Review* and *Meta-Analysis*. (If one is missing, use **Additional filters** → **Article type** to add it.)
- **Publication date:** *10 years*.
- **Additional filters** → **Age:** *Adult: 19+ years* (optional; explain the trade-off below).
- **Audience sees:** the count falls to the hundreds or fewer, and the filters appear above the results with an option to clear them.
- **Say:** *"Every filter needs a reason you can write down. Article type because ours is a treatment question. Ten years because practice has changed."*
- **Caution to voice:** age and species filters rely on indexing, so the very newest papers (not yet indexed) can be lost. Never use *Free full text* as a quality filter.

Step 5: Open one result (2 min), then Send to citation manager (30 s)

- **Do:** scan the titles aloud. Look for the large randomised trials and the course papers. Useful title words to watch for:
- *Restriction of Intravenous Fluid in ICU Patients with Septic Shock* (CLASSIC, Meyhoff 2022, *N Engl J Med*). Explain the name when it appears: CLASSIC stands for Conservative versus Liberal Approach to Fluid Therapy of Septic Shock in Intensive Care, a large randomised trial in 1,554 adults with septic shock in European ICUs. Restricting IV fluid did not change 90-day mortality.
- *Early Restrictive or Liberal Fluid Management for Sepsis-Induced Hypotension* (CLOVERS, Shapiro 2023, *N Engl J Med*)
- *Impact on fluid balance of an optimized restrictive strategy targeting non-resuscitative fluids…* (Paper 1, Boulet 2024, *Crit Care*)
- If one is not visible in the filtered list (it may be outside the filters, or ranked lower), say so honestly (that is a teaching point) and search its title words directly, e.g. `restriction intravenous fluid septic shock`.
- **Do:** open one record (CLASSIC is ideal).
- **Audience sees:** the abstract, authors, journal, the **MeSH terms** list (scroll down), **Similar articles**, **Cited by**, and full-text links (top right).
- **Say:** *"The MeSH terms list here is a free gift: it tells you the vocabulary to use in your own search."*

- **Show (30 s, if time allows):** on the PubMed home page, under **Explore**, open **MeSH Database** and type `bedsore`. The record shows the heading *Pressure Ulcer* and the words it collects. This is the other way to find a MeSH term (slide "How to find the right MeSH term").

Part B: Zotero (5 min)

Step 6: Send to citation manager → Zotero → Word (5 min)

- **Do (PubMed):** back on the results, tick 2–3 papers (e.g. CLASSIC and Paper 1), then choose **Send to** → **Citation manager** → **Create file**. A `.nbib` file downloads.
- **Do (Zotero):** in Zotero desktop, use **File** → **Import…** and choose the `.nbib` file. (Double-clicking the file often works too.)
- **Audience sees:** each paper appears as one entry with authors, title, journal, year, volume, pages, PMID and DOI, with nothing typed by hand.
- **Do (Word):** in the blank document type one sentence, e.g. *"Restricting IV fluid did not change 90-day mortality"*, then **Zotero tab** → **Add/Edit Citation**, type `Meyhoff` (the first author of CLASSIC) and press Enter. Then **Add/Edit Bibliography**. Then open **Document Preferences** and switch the style from a numbered (Vancouver-type) style to **APA** and back.
- **Audience sees:** `[1]` becomes `(Meyhoff et al., 2022)` and the reference list reformats itself. This is the "Citation styles" slide that came just before, done live.
- **Say:** *"Collect, export, import, cite, list. Never type a reference by hand, and never cite a paper you have not read."*
- **Participants:** everyone installed Zotero as Session 1 homework, so they can repeat the import on their own computer as you go.
- **Mention (do not demo):** the **Cite** button on each PubMed record (one reference); **Clipboard**; a free **My NCBI** account to save searches and get e-mail alerts (*Create alert*); the **Zotero Connector** browser button that saves a paper from a journal page in one click. Mendeley and EndNote work the same way.

Part C: Google Scholar (optional extra slide, 3 min)

Step 7: Google Scholar, for comparison

- **Do:** in the Scholar tab, search `restrictive fluid septic shock mortality`.
- **Audience sees:** a broad mix (journal articles, theses, reports, preprints), ranked by a hidden algorithm, with "Cited by N" under each result.
- **Do:** click **Cited by** under a large trial (e.g. CLASSIC), then **Search within citing articles** for `mortality`.
- **Say:** *"Scholar is excellent for following citations forward and for grey literature, but there is no MeSH, few filters, and the same search may give different results next month. Use PubMed for the search you report. Use Scholar to check you have not missed anything."*
- **Optional:** show the **Cite** (quote-mark icon) button and its export formats, such as EndNote or RefMan.

Close (30 seconds)

Ask: *"Which step reduced the hits most?"* Expected answer: combining the blocks with AND and the article-type filter.

After Part A, participants do Exercise 1 (Boolean search builder, 5 minutes). After Part B, continue with the slide "Citing vs copying: paraphrase safely".

Teaching points to land (say each once)

1. **Write down** the database, the date, the exact string, the filters and the number of hits, so the search can be repeated.
2. **OR inside, AND between**. Use NOT only with a reason.
3. **MeSH + free text** in every block, because the newest papers may not have MeSH yet.
4. **Filters need a reason**. Never filter for free full text.
5. **Screen by reading**: the computer narrows, people choose.
6. **Export, don't type** references.

If something goes wrong

Problem	What to do
No internet / PubMed slow	Switch immediately to the offline screenshots (or ask the second trainer to share them). Narrate the same steps. Do not troubleshoot live
A "Too many results" or error page	Paste the full string from the text file rather than typing
Zero results	A missing bracket or quote. Show Search details. It becomes a teaching moment
A known paper does not appear	Explain why (filters, indexing delay, different words) and search its title directly. This is a real-world lesson
Google Scholar shows a CAPTCHA	Do not try to solve it while sharing your screen. Skip to the screenshot of Scholar results
Running over time	Skip Step 7, use the fast route in Step 2, and show only the Zotero import (not Word) in Step 6
Zotero import fails or not installed	Show the screenshots of the import and the Word citation, and say "the same file works in Mendeley and EndNote"

After the demo

Participants copy the demo string on their sheet (Exercise 1). In Proposal builder 2 (2:20–2:55, 21:50–22:25) the trainer first re-runs this string and screens it live with the class (participants suggest terms and papers in the chat); then each participant runs their own string and saves their five references to Zotero. Trainers answer questions in the chat.